ISAAC LAB / NEWTON RESEARCH NOTES

Shared Mechanical Policies

for Cross-Embodiment Locomotion

Architecture, experimental evidence,
limitations, and research directions

7 October 2026

Current conclusion

A single mechanical-token attention policy reproduces fast locomotion on the three- and four-leg training robots after specialist distillation. This establishes source-control capability in the present setup. Frozen transfer tests find slow motion on five legs and rapid falls on simulated Unitree Go2/A1. A subsequent Go2 specialist learns the task, but solo and mixed-family attention imitation still fail on Go2. Mechanical understanding, reliable transfer and an original algorithmic contribution remain research questions.

Document status. Consolidated technical report of existing code and saved experiments. Proposed methods and experiments are labeled explicitly. The October 7 extensions add frozen target evaluations in Section 7 and new Go2 specialist/student training in Section 8, all with the existing training seed 42. No independent training-seed sweep was performed. The accompanying source package contains the result snapshots, vector figures, bibliography, and rebuild script.

Contents

1 Executive assessment **1**
1.1 Evidence and operating scope . . . 1

2 Problem definition and experiment contract **2**
2.1 One policy over a family of mechanical systems . . . 2
2.2 The current Ant benchmark . . . 2
2.3 Performance and accounting . . . 2

3 Mechanical representation **3**
3.1 Typed entities and physical relations . . . 3
3.2 Frames, scaling and normalization . . . 4
3.3 Supported scope and caching . . . 4

4 Actor architecture and attention **4**
4.1 Exact forward path . . . 4
4.2 What query, key and value mean here . . . 5
4.3 Attention-free and pooling comparisons . . . 5
4.4 Relation to URMA's routing mechanism . . . 6
4.5 Training components versus deployment components . . . 6

5 Specialist-imitation experiment **6**
5.1 Question and controlled changes . . . 6
5.2 Data collection and separation . . . 7
5.3 Body-balanced loss and fitting . . . 7
5.4 Rollout and implementation checks . . . 8

6 Results and what they establish **8**
6.1 Source control after distillation . . . 8
6.2 Action-fitting error . . . 10
6.3 Earlier PPO architecture and diversity screen . . . 10
6.4 Earlier inertial-feature ablation . . . 10
6.5 Interpretation limits . . . 11

7 Frozen transfer to additional robots: October 7 extension **11**

8 Learning a new family: Go2 specialist and distillation **12**
8.1 A competent reference under the declared task . . . 12
8.2 Controlled imitation comparison . . . 13

8.3 Frozen rollout results . . . 13

9 Capabilities, limitations and failure modes 14

10 Related work and novelty assessment 15

10.1 What the current ingredients already overlap . . . 15

10.2 Why a generic physics auxiliary objective is insufficient novelty . . . 16

10.3 Different types of possible contribution . . . 16

11 Proposed research directions 17

11.1 A: local mechanical interventions and structural transfer . . . 17

11.2 B: equivalent-description consistency . . . 18

11.3 Why the two requirements belong together . . . 18

12 Ordered experiment program 19

12.1 Phase 1: freeze and measure the current actors . . . 19

12.2 Phase 2: audit mechanics and conventions without retraining . . . 19

12.3 Phase 3: separate physics variation from template identity . . . 19

12.4 Phase 4: test one justified mechanism . . . 20

12.5 Phase 5: test new structures under a locked recipe . . . 20

12.6 Conditional later branches and stopping rules . . . 20

13 Recommended claim and immediate decision 21

A Complete static feature schema 22

B Source prototypes and split meaning 23

C Implementation and artifact map 24

C.1 Selected checkpoints and experiment records . . . 24

D Rebuilding and provenance 25

References 26

1 Executive assessment

The practical objective is to reuse a controller across mechanical systems with different numbers of bodies and actuators. The immediate question was more basic: why did our shared graph and attention policies move much more slowly than conventional specialist neural networks, even on their training robots?

The latest diagnostic changes the answer. Keeping the actor architecture and physical inputs intact, specialist imitation produces competent shared source control. The leading attention checkpoint reaches 5.566 and 5.032 m/s on the three- and four-leg robots, with 97.9% and 100% time-limit survival. Those speeds are 98.3% and 95.8% of the freshly evaluated teachers. The same attention architecture trained with PPO from scratch previously reached 1.276 and 2.346 m/s. The learning procedure therefore matters substantially in this setup.

The subsequent frozen transfer tests in Section 7 expose the remaining gap: both actors crawl on five-leg Ant and fail to sustain stance on the effort-controlled Unitree targets. Additional training families and the control interface need investigation before increasing network size.

The new-family learning experiment in Section 8 establishes a competent Go2 MLP reference, but small expert-state imitation error does not translate to stable student rollouts. This failure also occurs in the Go2-only fit, so interference with Ants cannot be its sole explanation. The immediate learning priority is startup and recovery-state coverage, with the existing architecture and matched extra-fitting controls.

The result supports three decisions:

1. Retain distilled attention as the working source-control baseline. Keep the attention-free mechanical graph as a comparison because it also produces fast gaits, although its survival is lower.
2. Test transfer and mechanical-feature use before another broad architecture sweep. A generic capacity objection is insufficient to explain the source gap.
3. Frame novelty around a specific, measurable property of mechanical reuse. Shared graphs, morphology-conditioned attention and specialist distillation all have close precedents.

Preferred research question

Can one controller respond usefully to real changes in a robot's mechanics, remain consistent when only the model's coordinate conventions change, and carry both properties to previously unseen articulated structures?

This question has practical applications: payload and motor changes, model import across different conventions, and reduced training effort for new robot structures. It also gives us falsifiable hypotheses. If direct conditioning and ordinary canonicalization already solve the problem, an elaborate new network is unnecessary. If richer mechanics contribute little under the current rich state observations, that is useful evidence about where the representation matters.

1.1 Evidence and operating scope

The experiment snapshot uses one teacher/student training seed and three evaluation reset seeds. At the user's request, additional independent-seed runs are outside the current plan; there is no seed-count gate on exploration. The rollout repetitions describe variation for these trained checkpoints, not the reliability of retraining the algorithm. Held-out bodies and disjoint demonstration splits remain essential to the questions being asked.

The current result is a research pilot, not a universal-robot guarantee. Its task is flat-ground forward

progress with effort control and rich simulator observations. The frozen transfer extension in Section 7 finds slow five-leg motion and failure on simulated Unitree targets. Those measurements are completed. Section 8 also completes the first new-family learning comparison; further representation experiments remain proposals.

2 Problem definition and experiment contract

2.1 One policy over a family of mechanical systems

Let $\mathcal{M}$ describe a robot's supported rigid-body mechanics: bodies, joints, actuators, collision geometry and relevant physical parameters. The simulator evolves a state according to

$$s_{t+1} = F_{\mathcal{M}}(s_t, u_t), \qquad o_t = O_{\mathcal{M}}(s_t), \qquad a_t = \pi_\theta(o_t, \mathcal{M}). \tag{1}$$

The physical control u_t is obtained by applying the environment's execution mapping to the actor's raw output a_t. In the current effort benchmark,

$$u_{t,k} = g_k(\mathcal{M})\ \mathrm{clip}(a_{t,k}, -1, 1), \tag{2}$$

where g_k is the declared gear/action-effort gain. A different robot can have a different number of actuator entries k, while sharing all actor weights.

Three distinct capabilities should be kept separate. *Interface support* means that a robot can be encoded and produces an appropriately sized action vector. *Source competence* means that training robots execute the task well. *Transfer* means that an unseen robot benefits without source retraining or with less target learning. The first does not imply either of the others.

2.2 The current Ant benchmark

The source pair consists of a radial three-leg Ant variant and the canonical four-leg Ant. The current family has two actuated joints per leg, giving six, eight or ten effort actions for three-, four- or five-leg bodies. Control is at 60 Hz. The objective rewards forward progress on flat terrain; it is not a commanded-speed or commanded-yaw tracking benchmark.

The richer mechanical policy observes body poses and velocities, joint state, previous raw actions, and incoming foot-link joint wrenches. Those wrenches are not direct measurements of foot-ground contact force. Availability of all these signals on physical hardware has not been demonstrated.

The earlier diverse population contains eight prototypes but only two source topologies. Its variations change scale, torso density, limb density, motor strength and damping. Appendix B lists the actual settings. The repeatedly inspected five-leg body is development data; it is not an untouched final test.

2.3 Performance and accounting

The saved speed metric is the existing completed-episode progress statistic. Survival is the fraction of episodes reaching the time limit, averaged over the evaluation reset runs. Speed must be read together with survival because fast partial behavior can coexist with instability. A useful source ratio is

$$\rho_b = \frac{v_{\mathrm{shared},b}}{v_{\mathrm{teacher},b}}, \tag{3}$$

for body b under the same task and observation/action contract. Near-zero or negative teacher speeds would make this ratio uninformative. No such issue arises for the current canonical teachers.

Our engineering source criterion is at least 90% of each competent teacher's speed, at least 95% survival, and no more than a two-percentage-point survival loss relative to that teacher. These are operating criteria, not optimality theorems. Teacher training, demonstration collection, student optimization and evaluation must be accounted for separately; the imitation study is not an equal-cost comparison to PPO from scratch.

3 Mechanical representation

3.1 Typed entities and physical relations

The exporter reads supported attributes from the finalized Newton model and builds a typed graph. Bodies, joints, DOFs and collision shapes are all explicit entities. Keeping these separate allows an unactuated body or collision shape to contribute information without requiring an action output for that entity.

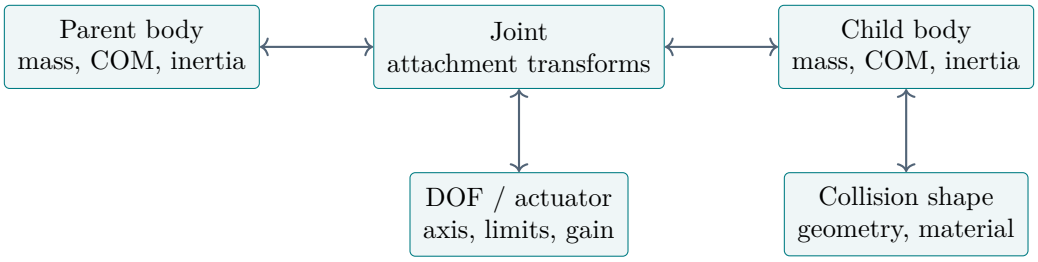


Figure 1: One illustrative mechanical subgraph. The implementation distinguishes the two directions of attachment relations. Permitted shape-contact relations are also represented; they describe eligibility, not the current contact schedule.

Each entity has 104 static slots and 23 dynamic slots, with inapplicable slots zeroed. A four-way kind indicator identifies the entity type. A separate 24-value global input contains 15 state values and nine context values: counts, total mass, control/simulation timesteps and gravity. The canonical source robots have 37 and 46 entity tokens, plus one global token.

Entity	Static examples	Current-state examples
Body	Mass; COM; full 3×3 inertia; dynamic/kinematic flags; root indicator	Root-relative pose; COM linear and body angular velocity; available incoming joint wrench
Joint	Type; parent/child attachment transforms; enable flag; DOF dimensions	State is mainly carried by the associated body and DOF tokens
DOF	Axis; angular flag; limits; armature; damping; friction; target mode/gains; effort and velocity limits; action gain	Normalized joint coordinate; scaled velocity; previous raw action on actuated DOFs
Shape	Primitive; local transform; scale; collision flags; margins; material coefficients; world-shape flag	No independent moving-shape state; attachment connects it to the body

Table 1: Information made available to the actor. Availability does not establish that the trained policy uses each field. Appendix A gives every static slot and its offset.

3.2 Frames, scaling and normalization

Body positions and orientations are expressed relative to the root link. Absolute body COM linear velocities and body angular velocities are expressed in the root axes; these are not velocities with root motion subtracted. Runtime body quaternions use $wxyz$, while exported Newton transforms use $xyzw$. The runtime pose encoding chooses a quaternion hemisphere to remove the $q/-q$ ambiguity. These conventions matter when comparing supposedly equivalent models.

Static channels undergo a fixed transform

$$\widetilde{m}_{i,r} = \operatorname{sign}(c_r m_{i,r}) \log(1 + |c_r m_{i,r}|)\,, \tag{4}$$

with inertia multiplied by 1000 and armature by 100 before compression; other static channel scales are one. Dynamic channels and the first 15 global values use empirical normalization. The remaining global context uses signed-log compression. In the successful imitation experiment, normalization is estimated from training demonstrations only and then frozen.

These choices make heterogeneous magnitudes easier to process, but they do not enforce dimensional analysis, unit invariance or a physical computation. A different unit convention must be mapped to the schema's declared units. Learned normalization must not absorb validation or target trajectories.

3.3 Supported scope and caching

The v1 schema targets rigid articulation trees with supported free, fixed, revolute and prismatic joints, analytic collision primitives and a flat ground plane. Mesh geometry, deformables, general closed loops, mimic/tendon mechanisms and coupled/custom actuation are not represented by this pilot contract. Some unsupported cases are rejected rather than silently approximated.

The description is cached per prototype. A reset-time or per-world change to mass, inertia or actuation requires a matching description refresh. Otherwise the policy receives false information about the simulated system. The relevant physical attributes belong in the representation; solver caches, rendering data and arbitrary identifiers do not belong there simply because they are fields on a Newton model.

4 Actor architecture and attention

4.1 Exact forward path

For entity i, concatenate compressed mechanics $\widetilde{m}_i$ and normalized dynamic state $\widetilde{x}_i$ and encode them with shared parameters:

$$h_i^{(0)} = \mathrm{ELU}(W_e[\widetilde{m}_i; \widetilde{x}_i] + b_e) \in \mathbb{R}^{64}. \tag{5}$$

A separate linear projection creates one global token. All tokens then pass through one local relation-aware attention block and two full-attention blocks. There are four heads, hence 16 values per head. Each block uses pre-normalization, a residual attention update, and a residual feed-forward update with width $64 \to 256 \to 64$ and GELU activation.

For actuated DOF k, the final token is decoded as

$$a_{t,k} = w_o^\top \,\mathrm{ELU}\left(W_d\,\mathrm{LN}(h_k^{(3)}) + b_d\right) + b_o. \tag{6}$$

The same decoder applies to every actuator. Padding is excluded from attention and masked from actions and PPO action-density calculations. No actor robot ID or token-position embedding is used. However, counts and geometry can still identify familiar templates, so absence of explicit IDs does not establish compositional generalization.

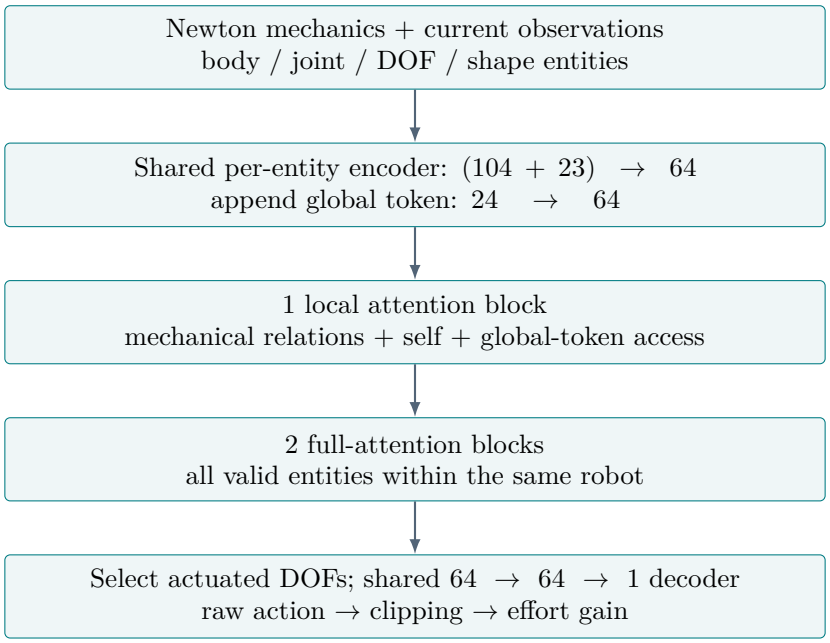


Figure 2: The deployed attention actor. Static and dynamic information mix at the first encoder. There is no temporal memory, predictive world model, or separately pretrained mechanical encoder in this checkpoint.

4.2 What query, key and value mean here

For one head in a block, let $\bar{h}_i = \mathrm{LN}(h_i)$ and compute

$$q_i = W_Q \bar{h}_i, \qquad k_j = W_K \bar{h}_j, \qquad v_j = W_V \bar{h}_j, \tag{7}$$

$$e_{ij} = \frac{q_i^\top k_j}{\sqrt{16}} + b_{r(i,j)} + A_{ij}, \qquad \alpha_{ij} = \frac{\exp(e_{ij})}{\sum_\ell \exp(e_{i\ell})}, \qquad c_i = \sum_j \alpha_{ij} v_j. \tag{8}$$

Here $b_{r(i,j)}$ is a learned head-specific relation bias and A_{ij} is zero for allowed senders and $-\infty$ otherwise. The local block allows mechanical neighbors, self and global-token connections. The next two blocks allow all valid tokens in that robot and retain relation biases.

The query specifies what a receiver currently seeks; the key supplies features against which that request is scored; the value contains the information that will be communicated. All three derive from mixed mechanics and state, so the routing can depend on motion as well as structure. Physical connectivity is known, but task-relevant information flow need not remain local or constant.

For example, a leg actuator could read distant body motion when correcting a whole-body disturbance. This is a capability of the network, not a measured interpretation of the learned heads. An attention map alone does not demonstrate causal physical reasoning. Attention also does not supply a mass matrix, solve contact constraints, or guarantee extrapolation to a longer kinematic chain.

4.3 Attention-free and pooling comparisons

The graph alternative replaces all three attention blocks. Its grouped local message has the schematic form

$$c_i^{(g)} = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} w_{r(i,j)}^{(g)} \phi(\bar{h}_j)^{(g)}, \qquad \bar{h}_{\mathrm{global}} = \frac{1}{N} \sum_{j \text{ valid}} \bar{h}_j. \tag{9}$$

A learned projection of the global mean supplements local messages. The four groups have 16 channels each, and the message MLP is $64 \to 64 \to 64$. Edge-type coefficients are learned but do not depend on a current receiver/sender query-key comparison. There are no dot-product attention calls or softmax routing weights in this path.

Variant	First block	Next two blocks	Parameters
Attention	Local relation attention	Full relation-biased attention	164,218
Graph	Typed messages + global mean	Same message operation	164,218
Hybrid pooling	Local relation attention	Learned scalar pooling/broadcast	147,628

Table 2: Current actor families. Counts include the shared scalar PPO exploration parameter. Matching parameter counts does not match FLOPs or isolate a single operator. The graph implementation uses dense tensors.

The hybrid pooling actor is not attention-free because its first block is still local attention. Nor is it a faithful URMA implementation. Both graph and attention retain individual actuator tokens; neither should be confused with the older joint-only graph and its 32-value pooled latent.

4.4 Relation to URMA's routing mechanism

URMA separately encodes joint descriptions and joint observations. Its encoder uses descriptor-derived softmax weights to route observation features into a shared latent; a core network and descriptor-conditioned per-joint decoding then produce actions [1]. This is not the standard pairwise token-to-token $QK^\top$ operation used in Equation (8). A useful intuition is that descriptions identify where a signal should be routed, while observations supply its changing content. Calling the description a "key" is only an analogy; it should not imply a missing conventional query projection in that encoder. Our policy instead retains all mechanical entities and lets mixed-state queries and keys choose their communication. Both designs accommodate variable input/output counts, but neither property guarantees successful control of any robot.

4.5 Training components versus deployment components

The PPO implementation uses separate flat MLP critics for the prototypes; only these critics receive prototype indices. A scalar log standard deviation supplies shared Gaussian exploration for the actor. The critics are not used at deployment. During imitation, frozen teacher means provide labels, no PPO critic is trained, and the actor's exploration parameter is unused. The deployed distilled checkpoint contains one shared actor, not a robot-specific teacher ensemble or a target adapter.

5 Specialist-imitation experiment

5.1 Question and controlled changes

The diagnostic asks whether the mechanical actors can represent the competent behavior already found by matched MLP specialists. Student weights start fresh; the students receive no copied teacher weights. Physical inputs, action mapping, task and actor architecture are preserved. The change is the learning procedure: fit teacher action means on collected source states, then execute the fitted student in the simulator.

The teachers are separate canonical MLP means with widths $[256, 128, 64]$ from the common PPO experiment. Their original optimization/exploration machinery was shared, so they should not be described as fully independent PPO learners. The fresh evaluations in this study establish the appropriate current reference.

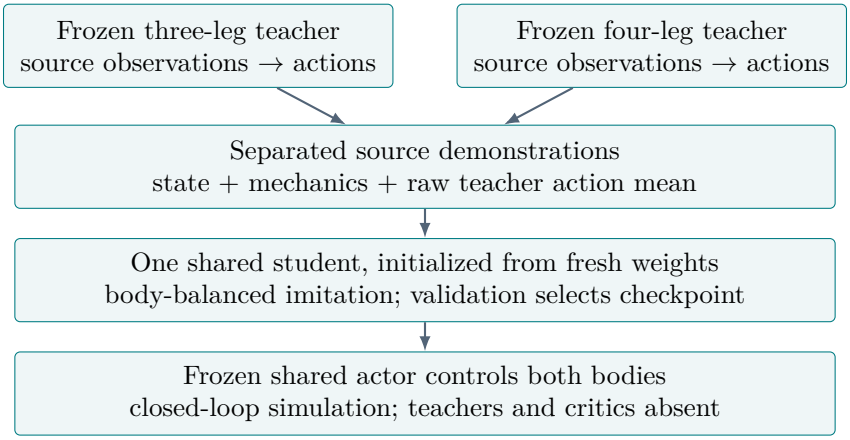


Figure 3: Training and deployment are distinct. Teacher action labels guide supervised fitting; teacher weights are not copied into the student and teachers are not selected at runtime. This experiment changes learning, not actor size.

5.2 Data collection and separation

For each of four collection reset seeds, run 64 worlds per body for 1,200 control steps and store every fourth pre-action state. Save raw deterministic teacher means, executed clipped/scaled efforts, physical descriptors and world/episode identities. The splits are:

Split	Reset seeds	States per body	Use
Training	1011, 2022	38,400	Gradients and normalization
Validation	3033	19,200	Checkpoint selection
Source-state test	4044	19,200	Action-error reporting

Table 3: Entire collection runs are separated. A source-state test is not an unseen-morphology test. No five-leg data enter this study.

The four collections consume 614,400 simulator transitions in total, including validation/test collection. The teachers previously used 16.384 million training transitions. Repeated presentations of stored examples add optimizer work, not new environment transitions.

5.3 Body-balanced loss and fitting

For a batch of B bodies with n_b examples and K_b valid actuators, the loss is

$$\mathcal{L}_{\mathrm{BC}} = \frac{1}{B}\sum_{b=1}^{B}\frac{1}{n_b}\sum_{t=1}^{n_b}\frac{1}{K_b}\sum_{k=1}^{K_b}\left(\pi_\theta(o_{b,t},\mathcal{M}_b)_k - a^*_{b,t,k}\right)^2. \tag{10}$$

Thus, a longer action vector does not automatically dominate training. Targets are raw teacher means, not clipped efforts: previous raw action is observed and raw action penalties also affect the task.

Each main fit runs 6,000 Adam steps with 128 sampled examples per body per step, or 768,000 sample presentations per body with replacement. The learning rate decays by cosine from 3×10^{-4} to 3×10^{-5}; gradient norm is limited to one. Normalizers are computed once from training data and frozen. Checkpoints are selected by minimum mean validation action MSE at step 1 and every 250 steps. All five main fits selected step 6,000.

The conditions are a shared attention student, a shared graph student, a separate graph for each body, and a separate-MLP-means imitation control. Each separate graph owns a full network; their combined parameter count is twice that of one shared graph. Separate versus joint fitting also changes normalization, so the comparison does not isolate gradient interference alone.

5.4 Rollout and implementation checks

Selected checkpoints are frozen and evaluated for 1,200 steps with 64 worlds per body and reset seeds 123, 456 and 789. These evaluation trajectories do not select checkpoints. The queue completed 28 jobs: four collections, six fits including a small memorization check, and 18 teacher/student evaluations.

The recorded checks cover body-balanced loss, padding gradients, raw-versus-clipped targets, collection-split rejection, checkpoint restoration and frozen normalization. Stored states reproduce teacher labels within 7.8×10^{-7} across the checked CPU/GPU paths. Descriptor comparison tolerates absolute 10^{-12} transform roundoff with zero relative tolerance; integer topology is exact. This addressed a 1.4×10^{-17} translation discrepancy without permitting a changed robot. A 64-state-per-body memorization check reached training RMSE 0.036/0.047 after 1,000 updates. It is a fitting check, not a new locomotion candidate.

6 Results and what they establish

6.1 Source control after distillation

	Three legs		Four legs	
Controller	m/s	Survival (%)	m/s	Survival (%)
Teacher MLPs	5.660	98.4	5.253	100.0
MLP imitation control	5.631	97.5	5.196	100.0
Graph, separate per body	5.625	97.5	5.055	95.8
Graph, shared imitation	5.500	94.0	5.013	96.9
Attention, shared imitation	5.566	97.9	5.032	100.0

Table 4: Fresh frozen-checkpoint evaluations from the October 7 diagnostic. Values are means over three evaluation reset runs. The table is generated from the included result snapshot rather than copied manually.

The attention student satisfies the numerical source criteria on both canonical bodies. The shared graph recovers 97.2% and 95.4% of teacher speed, but its three-leg survival is below 95%, and both bodies lose more than two percentage points against their teachers. Attention is therefore the current engineering baseline; the result does not establish universal architectural superiority.

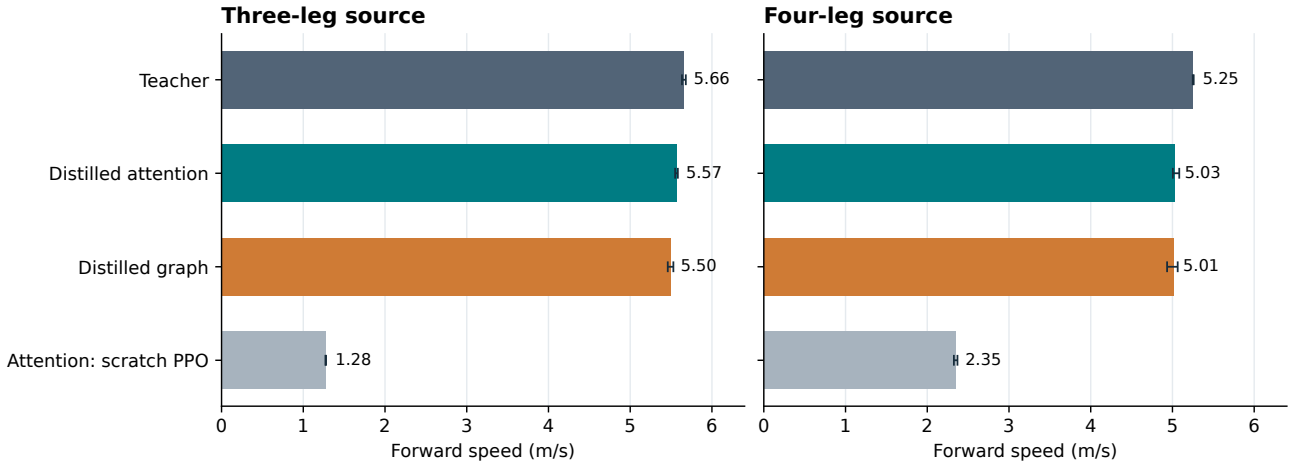


Figure 4: Source speed under the teacher, two distilled actors, and the earlier scratch-PPO attention policy. Whiskers show the minimum/maximum reset-run means, not confidence intervals over retraining. Distillation includes teacher and data collection costs, so this is a learning-procedure diagnostic, not an equal-cost algorithm ranking.

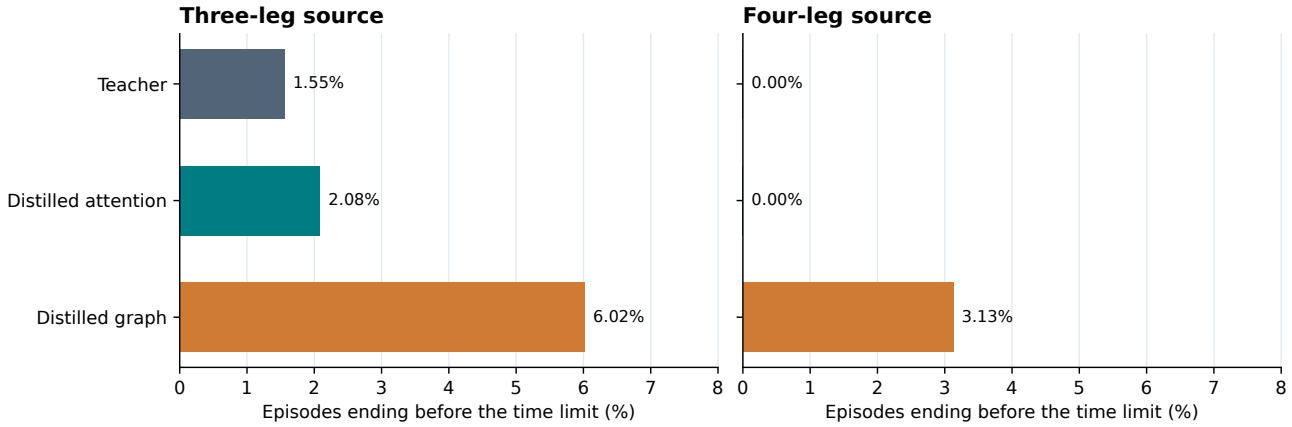


Figure 5: The complement of reported time-limit survival. Graph imitation retains fast motion but has more early-ending episodes. Zero observed early endings is not a guarantee of failure-free operation.

6.2 Action-fitting error

	Three legs		Four legs	
Student	Raw RMSE	Relative (%)	Raw RMSE	Relative (%)
MLP imitation	0.024	3.0	0.022	3.1
Shared graph	0.074	9.5	0.086	11.9
Shared attention	0.063	8.0	0.075	10.4

Table 5: Held-out source-state action errors. Relative RMSE divides by the training-label standard deviation; it is not a control success rate.

The shared attention student fits the actions better than the shared graph, while both remain less accurate than the MLP control. Closed-loop outcomes are still necessary: errors on teacher-visited states do not capture state-distribution drift or recovery after an unusual student action. Small action MSE does not guarantee stability.

6.3 Earlier PPO architecture and diversity screen

Sources	Actor	Three legs	Four legs	Five-leg development
2	Attention	1.276 / 99.5	2.346 / 90.0	0.726 / 100.0
2	Graph	1.920 / 100.0	2.480 / 99.0	1.289 / 100.0
2	Hybrid pooling	2.290 / 99.5	2.132 / 100.0	0.558 / 100.0
8	Attention	1.089 / 99.0	2.996 / 99.5	0.875 / 100.0
8	Graph	2.002 / 100.0	2.788 / 100.0	0.197 / 100.0
8	Hybrid pooling	1.253 / 74.6	3.110 / 100.0	0.451 / 100.0

Table 6: Earlier mechanical-policy PPO runs: speed in m/s / survival in percent. Each uses 500 updates, 16.384 million aggregate transitions and training seed 42. These five-leg results do not describe the later distilled checkpoints.

The graph improves some source measurements under PPO but has no consistent transfer advantage. Its five-leg speed falls sharply in the eight-prototype run despite 100% survival. Because aggregate experience is fixed, eight prototypes receive one quarter the per-prototype exposure of the two-prototype run. More diversity and less per-body experience change together, so this is not an isolated causal test of diversity.

An earlier joint-only graph was also extended from 500 to 2,500 updates. Solo three-/four-leg speeds rose from 1.502/2.544 to 1.880/3.895 m/s; shared speeds rose from 1.732/2.149 to 2.308/3.310 m/s. Longer training helped that architecture, with smaller late gains, but it remained a different representation from the current mechanical actor. Those results neither establish nor rule out the value of longer training for the new learning recipe.

6.4 Earlier inertial-feature ablation

For eight-source attention/PPO, removing mass, COM, inertia, armature and total mass inputs changed canonical three-/four-leg and development five-leg speeds from 1.089/2.996/0.875 to 1.895/2.094/0.170 m/s. Three legs improved while four and five legs worsened. This is mixed evidence

for the utility of those inputs, not proof of a learned inertial model. The ablation retrains PPO, changes learning dynamics, and still supplies geometry and dynamic state that can reveal physical properties. It has not been repeated on the successful distilled baseline.

6.5 Interpretation limits

The positive source result refutes a categorical claim that the current shared architecture cannot express the fast source behaviors. It does not distinguish exploration, credit assignment, normalization, optimizer behavior and cross-body interference: imitation changes several of these together. Nor does it show that an unseen body is controlled using its mechanics rather than template recognition. Older specialists at roughly 6.25/5.37 m/s used a different contract; the freshly evaluated teachers in Table 4 are the appropriate local reference.

7 Frozen transfer to additional robots: October 7 extension

The same source-selected attention and graph checkpoints were applied to the five-leg development Ant and to Unitree Go2 and A1 USD assets. No target data entered fitting, normalization or checkpoint selection. The actor, normalizers and output decoder were unchanged; the graph supplies ten actions for five-leg Ant and twelve for both Unitree models. This extends the accepted interface, but does not establish successful behavior on those robots.

Physical and evaluation contract. Each target/controller uses reset seeds 123/456/789, 64 worlds per run and 1,200 control steps at 60 Hz, with a 16-second episode limit and early resets. These are reset repetitions, not new training seeds. Speeds in Table 7 divide completed-episode progress by completed-episode time, then average the three run-level ratios. Survival is the fraction of completed episodes reaching the time limit. Mean episode duration is essential for interpreting a policy that repeatedly falls; negative speeds during those falls are not a backward locomotion gait. The data snapshot also contains progress over the entire rollout, including unfinished episodes, and action saturation rates.

Target	Actor	Speed (m/s)	Survival (%)	Episode (s)
Five-leg Ant	Attention	0.245	100.0	16.000
Five-leg Ant	Graph	0.176	100.0	16.000
Unitree Go2	Attention	-0.085	0.0	0.212
Unitree Go2	Graph	-0.058	0.0	0.231
Unitree A1	Attention	-0.002	0.0	0.207
Unitree A1	Graph	-0.000	0.0	0.223

Table 7: Frozen distilled actors on development targets. Unitree rows use the explicit simulated effort-control conversion below, not stock hardware drives.

Unitree geometry, masses, joint layout, nominal posture and configured effort limits are retained. Their stock DC motor/position controllers are replaced by ideal effort control with 0.1 N m s/rad damping, matching the Ant's damping convention. This is a declared model conversion, not an exact stock-motor test. The existing descriptor has four drive-mode slots and rejects Newton's fifth, pure-effort mode; it can encode the velocity-drive mode used for this damping. No field was silently remapped or added to the trained network.

Unitree simulation uses 480 Hz integration while preserving 60 Hz policy updates. The original 120 Hz integration produced non-finite Go2 observations under large efforts. Constraint/contact capacities were increased to 2,048/256 per world after a capacity overflow in an initial probe. Failed probes are retained in the logs and excluded from scores. Ant integration remains 120 Hz. Both the physics settings and descriptor values are recorded per run.

The standing sanity check uses nominal-posture PD effort feedback at every Unitree physics step, with gains 25 and 0.5. It is an environment check, not a trained locomotion specialist or an equal-controller comparison. Go2 reaches 100% survival with this controller. A1 reaches only 16.6% with the same gains; a separate follow-up raising only its standing feedback damping to 2.0 reaches 100% survival over 64 worlds at reset 123. The learned-policy plant and weights are unchanged. This follow-up was prompted by the failed standing check and is not part of the original three-reset matrix. Neither target had a matched trained locomotion specialist in this frozen experiment; the later Go2 reference is established in Section 8.

Interpretation

Source imitation solves the earlier source-performance gap but does not solve transfer. The five-leg robot largely remains above its fall-height threshold while moving slowly. Both shared actors fall rapidly on the new quadruped family. The result does not isolate training coverage, mechanical reasoning, coordinate conventions, rich-state normalization or effort-control semantics as the cause. It motivates a controlled new-family learning comparison before another increase in network size.

The targets are now development data. The reserved five-leg parameter combinations remain untouched. The new evaluation tool is `scripts/tools/evaluate_mechanical_transfer.py`; raw reports, failed setup probes, descriptor tensors, fixed-world videos and a gallery are under `logs/shared_ant_graph/other_robots_20261007/`. The portable report includes `data/transfer_summary.json`, with the frozen protocol and compact reports.

8 Learning a new family: Go2 specialist and distillation

This extension separates learning the new robot from sharing its controller with the Ant family. Go2 becomes a training body. A1 and five-leg Ant remain development transfer probes, with no demonstrations or normalization samples used for either fit. Previously inspected development bodies are not an untouched confirmation suite.

8.1 A competent reference under the declared task

A conventional Go2 MLP specialist uses the same rich observations, ideal effort mapping, reset distribution, reward and 480 Hz physics/60 Hz control as the frozen transfer study. The actor has layers 256–128–64 and 124,621 parameters. It uses 1,024 worlds, 32-step PPO rollouts and seed 42. The first 500 updates (16.384 million transitions) produce stable standing but essentially zero forward speed. This fails the predeclared collection gate: mean speed at least 1 m/s and mean time-limit survival at least 95%.

A documented continuation adds 1,000 updates with the same weights, optimizer, normalizers and training configuration, restarting the simulator rollout state. At 1,500 total updates (49.152 million transitions), the frozen final checkpoint reaches 9.921 m/s and 100% survival across reset seeds 123/456/789. This establishes learnability of the simulated task. It is not a prediction of real Go2 performance: the converted model omits the stock motor torque–speed curve, and the objective maximizes forward progress rather

than tracking a bounded speed command. Longer training helped this specialist; the result does not establish that duration alone repairs the shared student's learning problem.

8.2 Controlled imitation comparison

The two students are fresh copies of the existing 164,218-parameter attention actor, each initialized with seed 42. One fits Go2 only; the other fits Ant3, Ant4 and Go2. Architecture, token schema, action decoder and training objective are unchanged. Four Go2 collections use 64 worlds, 1,200 control steps and sampling every fourth step: 307,200 new simulator transitions in total. Seeds 1011/2022 supply training data, 3033 validation and 4044 test data. Each body has 38,400 training rows and 19,200 rows per held-out split. Existing Ant collections are reused without changing their 120 Hz physics. Both families retain 60 Hz actions.

Stored graph layouts reproduce the live mechanical encoder exactly on checked observations. Descriptor snapshots, teacher hashes, dependency versions and observation widths are verified before fitting. This permits offline mixed-body batches without forcing Ant and Go2 to share a simulator integration timestep.

Each fit uses 6,000 Adam updates, 128 samples per body per update, a learning rate decaying from 3×10^{-4} to 3×10^{-5}, and equal body weighting of valid-DOF raw action MSE. The selected checkpoint minimizes source validation MSE, not rollout performance. Normalization is estimated only from that condition's training demonstrations and then frozen. Both conditions therefore have 768,000 Go2 sample presentations, but the joint condition has three times as many total presentations and different normalization statistics. It is a useful system comparison, not an isolated gradient-interference ablation or an equal-total-compute comparison.

8.3 Frozen rollout results

Robot	Specialist	Go2-only attention	Shared Ant+Go2
Three-leg Ant	5.660 / 98.4	—	5.467 / 89.8
Four-leg Ant	5.253 / 100.0	—	5.086 / 96.4
Go2 (new source)	9.921 / 100.0	0.804 / 0.0	-0.003 / 0.0
A1 (development)	—	0.031 / 0.0	0.233 / 0.0
Five-leg Ant (development)	—	—	0.131 / 97.4

Table 8: Speed (m/s) / 16-second survival (%). Each cell averages three reset runs, 64 worlds per body and 1,200 steps. Dashes denote unmeasured cells or missing matched references. Ant evaluation preserves the canonical paired-source layout; the Go2 teacher is the longer-trained reference above. These are fixed trained policies, not independent training replicates.

The Go2-only student has a held-out raw-action RMSE of approximately 0.018, yet fails to sustain locomotion. Small average error on expert trajectories is insufficient to establish closed-loop control. Because this failure occurs without Ant sharing, interference between embodiments cannot be its sole cause. Neither architecture capacity nor the proposed mechanical representation has been isolated as the cause by this comparison.

The mixed fit retains Ant speeds of 5.467/5.086 m/s, but Ant3/Ant4 survival falls to 89.81%/96.35%, versus 97.9%/100% for the earlier Ant-only attention checkpoint. Both fail the full source acceptance rule: Ant3 misses 95% survival, and Ant4 loses more than two percentage points relative to its teacher.

Joint training was fresh, with Ant replay throughout; this is not a sequential fine-tuning/forgetting experiment. Go2-only and shared Go2 episodes last only 0.680 and 0.212 seconds, respectively. A1 still falls rapidly and five-leg progress remains 0.131 m/s. The new shared checkpoint therefore does not replace the Ant-only incumbent.

Fresh trajectory diagnostic. A post-matrix check uses reset 123, 64 worlds and 1,200 steps per condition, executing either the teacher or student while evaluating both actors' actions. All weights and normalizers remain frozen. For Go2-only imitation, RMSE is 0.01950 on teacher-visited states and 0.27446 on student-visited states. The shared actor gives 0.04397 and 0.24865, respectively. On teacher trajectories, Go2-only error in the first 0.25 seconds is 0.04778, versus about 0.017 after one second. This supports examining startup coverage and state-distribution drift; it does not isolate a unique cause. Different behavior policies visit different states, and expert actions on fallen student states are not guaranteed optimal recovery labels. These four diagnostic rollouts add 307,200 simulator transitions, with no fitting or checkpoint selection. See `data/go2_diagnostics.json`.

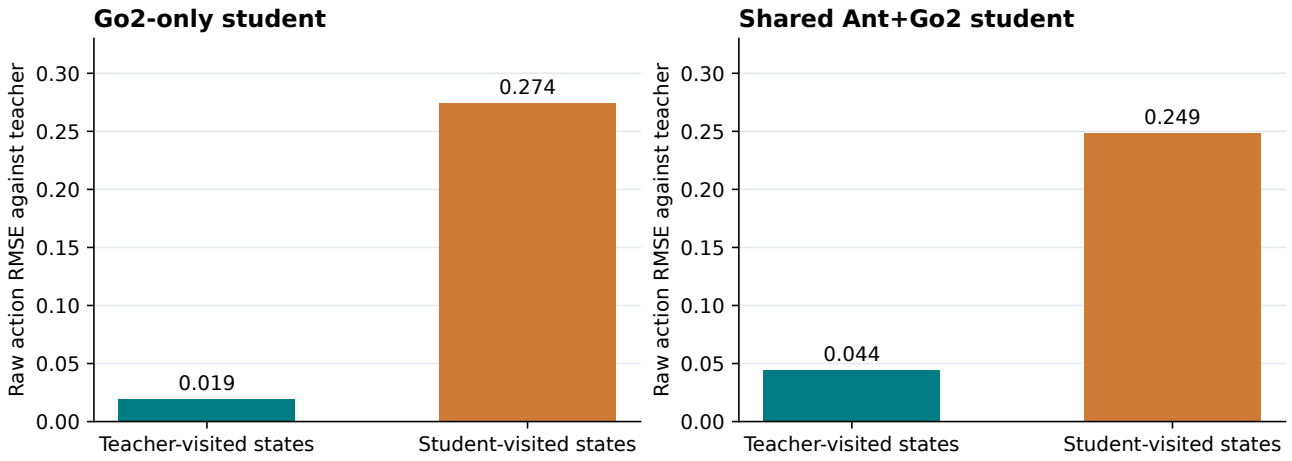


Figure 6: The small action errors on fresh expert trajectories do not persist on trajectories controlled by the students. RMSE is in dimensionless raw-action units. These are post-matrix diagnostics for the fixed checkpoints, not causal proof that dataset coverage alone explains the failures.

The next learning intervention should explicitly cover startup and student-visited recovery states, retaining Ant replay. Compare fitting augmented demonstrations with an equal-update continuation on the original data to distinguish additional state coverage from simply adding optimizer steps. Teacher actions on arbitrary fallen states are not guaranteed recovery targets; report expert recoverability and any assisted-rollout schedule. Preserve the current checkpoints, keep A1 out of fitting, and select checkpoints using source validation. This is a proposed repair of imitation learning, not a new-algorithm claim or evidence of universal transfer.

The complete protocol, checkpoints, raw reports and fixed-world videos are under `logs/shared_ant_graph/go2_learning_20261007/`. Tools are `scripts/tools/train_mechanical_go2.py` and `scripts/tools/distill_mechanical_cross_family.py`; the portable result snapshot is `data/go2_learning_summary.json`.

9 Capabilities, limitations and failure modes

Question	Current evidence or gap
One policy, several action counts	Demonstrated source control on six- and eight-action Ants; the interface accepts ten-action Ant and twelve-action Unitree variants.
Unseen morphology	Frozen distilled actors crawl on five legs and fail on the simulated Unitree targets; see Section 7.
Mechanical conditioning	Rich fields are exported. Their compositional use by the successful actor is unproven.
Embodiment diversity	Eight earlier prototypes still mean two source topologies; the successful distilled actor sees only the canonical pair. Adding Go2 has been tested, but the mixed fit is not competent on all three sources.
Physical-description consistency	Token permutations and padding are tested. Arbitrary body frames, axis signs and fixed-link decomposition have no guarantee.
Sensing and hidden dynamics	Current state is simulator-rich. No history or online system identification addresses uncertain mechanics or missing sensors.
Model coverage	General constraints, coupled actuators, deformables and mesh geometry require new schema/control support.
Practical compute	Actor parameter counts are known. Batch-one latency, control deadline headroom and broad scaling are not characterized.
Task coverage	Forward progress only. Command following, terrain, manipulation and hardware deployment have separate requirements.
Transfer efficiency	Earlier target-adapter pilots motivate reuse; the distilled mechanical actor has not established target-sample savings or a frozen-core benefit.

The strongest risks are not merely model size. Template shortcuts can make a mechanics-rich network appear physically informed. Cached descriptions can become stale under randomization. Coordinate-dependent numbers can make equivalent robots look different. Teacher-state coverage can leave the student unprepared for recovery. An implementation that accepts a graph does not resolve any of these issues automatically.

Attention is a useful flexible communication mechanism, but there is no current reason to treat it as mandatory. Conditional parameters, deterministic mechanical canonicalization, better data coverage or an improved training objective may be more effective than adding heads or layers. A change should address a measured failure and retain a matched simple control.

10 Related work and novelty assessment

10.1 What the current ingredients already overlap

Structured graph control is established by NerveNet [2]. Morphology-conditioned Transformer control and contextual modulation appear in MetaMorph and ModuMorph [3, 4]. URMA supplies a particularly relevant shared locomotion architecture [1]. Our explicit body, DOF and shape representation is a design choice to evaluate; a richer descriptor list alone is not a demonstrated research contribution.

The closest precedent to the successful learning recipe is UniLegs [5]: specialist locomotion policies are distilled into a shared Transformer. Its proprioceptive, joint-target setting differs from our richer

body/wrench observations and effort actions. These differences prevent a direct performance ranking and require task-matched tests. Its normalized-reward results cannot be compared numerically with our teacher-normalized speeds. Our present finding validates this implementation's source capability; it does not introduce specialist-to-Transformer distillation.

Reference family	Relevant overlap	Implication
NerveNet; MetaMorph; ModuMorph	Graph sharing, morphology conditioning and attention	These are baselines, not stand-alone novelty claims.
URMA	Shared locomotion with variable input/output counts	Variable-size control requires an empirical advantage beyond interface generality.
UniLegs	Specialist-to-shared-policy distillation	Our successful recipe has a close prior precedent.
Embedding Morphology into Transformers [6]	Joint tokens, topology-biased attention and attribute conditioning	Combining graph relations and descriptors with attention is also established.
Graph physics; QWM [7, 8]	Learned physical dynamics and morphology-conditioned world models	Adding a predictor is not sufficient.
CMIRL; GraphOp-WM [9, 10]	Mechanical interventions, structured morphology pathways and paired-morphology supervision	Counterfactual mechanics training needs a narrower distinction.
MS-PPO [11]	Graph policies respecting morphological symmetry	Equivariance alone is not a new contribution.

Table 10: Targeted primary-source comparison as of the report date. We have not reproduced these methods under a common benchmark.

10.2 Why a generic physics auxiliary objective is insufficient novelty

Graph networks have long been used for physical inference and control [7]. QWM conditions a quadrupedal world model on morphology and uses the learned dynamics for policy learning [8]. CMIRL separates morphology and shared-mechanism pathways and uses simulator morphology interventions [9]. GraphOp-WM combines attributed body graphs, structured operators and paired-morphology supervision, with controlled parameter-composition splits [10]. Consequently, neither "mechanics plus a world model" nor "learn from counterfactual physical changes" is a defensible broad novelty claim for our next experiment.

A more specific possibility is that *local mechanical effects learned on source structures transfer to different articulated topologies*, while physical control remains insensitive to arbitrary description conventions. That claim would require a defined mechanism, structural tests and matched alternatives; it is not established by our current source results.

10.3 Different types of possible contribution

An algorithmic contribution needs a new mechanism with a controlled benefit over stronger existing alternatives. An empirical contribution could identify when mechanical context helps, when it is redundant with state, or when optimization prevents its use. A benchmark/schema contribution could expose failures caused by equivalent model descriptions and supply validated transformation pairs. The current Newton exporter and experiment tooling are enabling infrastructure, not evidence that any of those contributions has already been achieved.

The literature check is targeted rather than exhaustive. In particular, description-frame consistency must be distinguished from morphological symmetry: the former can arise for an asymmetric robot because coordinate frames are a modeling choice. Existing symmetry-aware locomotion methods remain important comparators [11]. No first-of-kind claim is made for the proposed combination.

11 Proposed research directions

Proposed work: not implemented or evaluated

The two directions below are hypotheses to test using the source-competent actor. The existing checkpoint has no predictive auxiliary head or enforced local-frame equivariance. A successful source gait is a starting point, not evidence for either proposed property.

11.1 A: local mechanical interventions and structural transfer

The first hypothesis is that learning the effect of a local physical change can encourage a representation that transfers beyond recognizing source templates. Begin with fixed topology and geometry while changing valid payload/inertia and actuator parameters. Hold out parameter combinations. If the existing direct-conditioned actor already handles them, retain that simpler method.

For a matched simulator state s and physical control u, a pure inertial intervention gives

$$\Delta y = y(F_{\mathcal{M}'}(s,u)) - y(F_{\mathcal{M}}(s,u))\,, \tag{11}$$

where y extracts corresponding body/joint velocity features in consistent coordinates. Each pair has the same topology, so its correspondence is explicit. Train on many such local pairs from source structures, then test whether the learned relation transfers to a new structure. Do not define a naive elementwise subtraction between states with different numbers of joints.

For motor-strength changes, holding normalized action a fixed intentionally changes $u = C_{\mathcal{M}}(a)$ through the execution mapping. That is a different paired experiment and must be labeled accordingly. For inertial changes, hold applied effort fixed. Density interventions must update mass and inertia together; payload changes must produce a physically realizable COM and inertia. Contact state, constraint caches and derived wrench observations must be handled consistently when cloning and modifying a simulator state.

A possible auxiliary objective is

$$\mathcal{L}_{\text{paired}} = \|D_\theta(s,u,\mathcal{M}') - D_\theta(s,u,\mathcal{M}) - \Delta y\|_W^2\,, \qquad \mathcal{L} = \mathcal{L}_{\text{BC}} + \lambda \mathcal{L}_{\text{paired}}. \tag{12}$$

W declares feature scales rather than allowing the largest physical units to dominate. This is a candidate implementation, not a completed method. Compare it with ordinary next-state prediction using the same transitions and a matched prediction head. Dynamics prediction can improve without improving control, so held-out closed-loop performance is the decisive measurement.

Controls. Use direct-conditioned imitation, ordinary prediction plus imitation, and paired prediction plus imitation. Match teacher access, actor/head capacity, source data and simulator queries; count both branches of each paired simulation. Teacher labels must be generated by competent policies for the actual physical variants. Canonical teacher actions are not automatically expert labels for a heavier or weaker robot. Strict zero-shot tests allow no target dynamics queries in source training or checkpoint selection.

Success and falsification. Success is better control/survival on locked parameter combinations and new structures, with correct mechanical context materially useful. If only prediction improves, if matched-data controls erase the gain, or if removing inertial/actuator information preserves the result, the proposed mechanism has not earned its claimed role. Structural reuse, rather than counterfactual supervision by itself, is the proposed distinction from the nearby work discussed in Section 10.

11.2 B: equivalent-description consistency

The second hypothesis is that mechanically equivalent descriptions should lead to equivalent physical control. Let T change model coordinates without changing the robot, and let T_a map actions consistently. The desired property is

$$\pi_\theta(T_s s, T_{\mathcal{M}}\mathcal{M}) \approx T_a \pi_\theta(s, \mathcal{M}). \tag{13}$$

When T_a is nontrivial this is equivariance, not literal equality of action numbers. Our tested token-permutation property is only one special case.

For a scalar joint-axis reversal, an illustrative consistent mapping is

$$q' = -q, \quad \dot{q}' = -\dot{q}, \quad [q'_{\min}, q'_{\max}] = [-q_{\max}, -q_{\min}], \quad u' = -u. \tag{14}$$

The joint axis, reference configuration, previous action and other directional quantities must also transform consistently. For a local body-frame rotation, poses, attachment transforms, axes, COM vectors, inertia tensors and wrench components require corresponding transformations. Transforming only a few input vectors would describe a different physical system.

First verify generated pairs through forward kinematics and short dynamics checks under mapped actions. Then measure action disagreement after mapping both outputs to a common physical convention, followed by rollout speed and survival. This separates an incorrect transformation generator from a policy inconsistency.

Candidate remedies. Compare deterministic canonicalization, consistent data augmentation, and a learned encoder or consistency penalty enforcing Equation (13). A canonical frame cannot simply be assumed to be unique for symmetric or degenerate geometry; the convention and ambiguous cases must be explicit. A possible training penalty is

$$\mathcal{L}_{\text{equiv}} = \mathbb{E}_{s,\mathcal{M},T}\left[\left\|\pi_\theta(T_s s, T_{\mathcal{M}}\mathcal{M}) - T_a \pi_\theta(s, \mathcal{M})\right\|^2\right]. \tag{15}$$

This penalty alone admits unhelpful constant policies, so it must be paired with competent task/imitation performance and tests of sensitivity to real mechanics.

Fixed-link merging/splitting is a later, harder test. Combined spatial inertia, collision surfaces and constraints must remain equivalent, and numerical solver effects can break an assumed equivalence. Duplicating collision shapes is not a valid shortcut because it can change contacts. Start with exact coordinate re-expressions rather than ambiguous model edits.

Success and falsification. Success is lower physical-action/rollout discrepancy on equivalent descriptions, without losing source competence or useful response to actual physical changes. If ordinary canonicalization solves the issue, that is the preferred remedy. If the baseline is already sufficiently consistent, a new network is not justified. A contribution may lie in a validated schema and benchmark rather than a more complicated actor.

11.3 Why the two requirements belong together

A representation should ignore arbitrary conventions while retaining information that changes dynamics. Insensitivity to everything would pass a superficial invariance test but be useless for adapting to a heavier

link. Sensitivity to every descriptor perturbation would react to coordinate choices that do not change the robot. Evaluating both properties, and then structural transfer, offers a stronger test of reusable mechanics than either input ablation or attention visualization alone.

12 Ordered experiment program

The following program is conditional. It uses the current trained checkpoints first and introduces new training only when a measurement identifies a specific gap. No independent-seed sweep is included.

12.1 Phase 1: freeze and measure the current actors

Completed in the transfer extension. Section 7 records the target measurements; the following paragraph preserves the stage's protocol. Freeze the distilled attention and graph checkpoints, normalization and selection rule. Evaluate canonical sources and the existing five-leg development body with matched resets. Record speed, survival, action saturation and fixed-reset videos. The canonical evaluations already exist; repeat them only as needed to check a new evaluation path, rather than spending compute on redundant measurements.

Decision: does successful source imitation transfer at all? A negative five-leg result motivates representation/coverage work. A positive result is a development finding, not an untouched structural test or evidence of arbitrary robot control.

12.2 Phase 2: audit mechanics and conventions without retraining

Construct bounded, physically valid fixed-topology variants. For the same actual simulated variant, compare correct descriptions with deliberately mismatched descriptions that represent other valid variants. Keep observations and all non-target fields consistent. Separately apply validated equivalent-coordinate transformations and map the actions back.

Decision: are physical fields useful, ignored, or associated with brittle out-of-distribution behavior? Inference-time swaps are sensitivity probes, not conclusive causal evidence, because the mismatch itself creates distribution shift. A small action change can be appropriate under a robust policy; require a control-relevant effect rather than rewarding sensitivity for its own sake.

12.3 Phase 3: separate physics variation from template identity

Build a small factorial source population varying payload/inertia and motor strength while geometry is held fixed. Define held-out combinations before training. Validate physical realizability and descriptor refresh, then obtain competent per-variant teacher coverage. Train full-mechanics and geometry-only or masked-mechanics actors on identical observations, labels and sample counts.

Decision: does explicit mechanics improve control beyond identifying a template? Rich dynamic observations may still reveal physical properties, so a negative ablation limits the claim in this sensing/task regime, not in every possible partially observed setting. If variants lack competent teachers, resolve that reference failure before attributing poor students to sharing.

12.4 Phase 4: test one justified mechanism

Choose direction A when composition fails despite informative mechanical inputs. Choose direction B when equivalent descriptions cause meaningful discrepancies. Retain a direct-conditioned or canonicalized control. Change one principal mechanism at a time and compare data, capacity and compute explicitly.

If student state drift is the main failure, collect teacher labels on student-visited/recovery states, following the motivation of dataset aggregation methods such as DAgger [12]. This is a targeted repair, not a novelty claim. Imitation-initialized PPO is justified by a measured teacher ceiling or task mismatch; monitor whether early updates destroy the competent gait.

12.5 Phase 5: test new structures under a locked recipe

After development choices are fixed, evaluate a broader family-held-out suite: new leg counts, link counts, branching patterns, and eventually other supported body families. Separate parameter interpolation, parameter extrapolation, unseen combinations and structural change. If five-leg robots enter training, the old five-leg development result becomes within-topology evidence.

Compare against a credible shared morphology-conditioned/distilled baseline under the same inputs, control semantics, teachers and budgets. A port with substantial changes should be called a literature-aligned baseline rather than an exact reproduction. Use target specialists only for isolated scoring; their trajectories and model-selection signals must not leak into a zero-shot actor.

Report per-body results, minimum/median teacher-normalized speed, survival and failures. A sufficiently broad suite is needed before a lower-decile statistic is meaningful. Preserve both fixed-total-budget and fixed-per-body-budget comparisons when growing populations; otherwise training exposure and diversity remain confounded.

12.6 Conditional later branches and stopping rules

- **Longer training/capacity:** allocate it when curves or fitting errors motivate a test; retain fixed checkpoints and matched-compute controls. More layers or duration are remedies, not research contributions by themselves.
- **Command following:** introduce a separately versioned task with speed/yaw commands, feasible-command coverage and retrained references.
- **Memory and sensing:** compare rich state with restricted sensors; add history when hidden mechanics, delay or state ambiguity justify it.
- **Adaptation:** compare frozen zero-shot, adapters, partial/full fine-tuning and scratch at matched target budgets. Include random-frozen-core controls before claiming a reusable frozen core.
- **Compute:** profile batch-one latency and training throughput. Count teacher cost, simulation branches and optimizer steps separately. Existing 500-update PPO screens cost 16.384 million transitions each; their runtime is not a reliable forecast for a larger structural study.

Each run should record its hypothesis, one principal change, fixed controls, source/target split, action/sensor contract, budgets, checkpoint selection, success criterion, measured result and next decision. Retire a branch when its claimed mechanism fails matched controls. Keep a useful simple solution even when it weakens the intended novelty story.

13 Recommended claim and immediate decision

The statement justified now is: *in this Newton/Isaac Lab Ant experiment, specialist imitation enables a shared mechanical-token actor to recover most of the competent source speed, with attention giving better survival than the matched graph student.* The experiment shows source representability and a strong dependence on learning procedure. It does not establish universal control, mechanical reasoning or a new distillation algorithm.

Frozen development transfer has exposed a substantial gap, and the new-family specialist/shared-imitation comparison is now complete. The Go2 teacher succeeds while both attention students fail. The immediate learning experiment should compare startup/recovery-state coverage against equal extra fitting on the original data, retaining Ant replay. Controlled physical-description probes and a locked held-out morphology remain necessary for a transfer claim. The proposed research contribution is conditional: demonstrate that a specific representation or objective preserves equivalent physical control across description conventions while learning useful mechanical responses that transfer to unseen structures. If the measured gap can be solved by deterministic canonicalization or ordinary direct conditioning, use that result to narrow the project rather than forcing a more complex architecture.

A Complete static feature schema

The table below is generated from the implementation's `STATIC_FIELDS` dictionary. Offsets are zero-based and inclusive. A field is populated only for its applicable entity type. The full width is 104; padding adds tokens, not feature channels.

Field	Width	Offsets
`kind`	4	0–3
`body_mass`	1	4–4
`body_com`	3	5–7
`body_inertia`	9	8–16
`body_flags`	2	17–18
`root_body`	1	19–19
`joint_type`	8	20–27
`joint_X_p`	7	28–34
`joint_X_c`	7	35–41
`joint_enabled`	1	42–42
`joint_dof_dim`	2	43–44
`joint_axis`	3	45–47
`angular_dof`	1	48–48
`joint_armature`	1	49–49
`joint_target_mode`	4	50–53
`joint_target_ke`	1	54–54
`joint_target_kd`	1	55–55
`joint_damping`	1	56–56
`joint_effort_limit`	1	57–57
`joint_velocity_limit`	1	58–58
`joint_friction`	1	59–59
`joint_limit_lower`	1	60–60
`joint_limit_upper`	1	61–61
`joint_limit_ke`	1	62–62
`joint_limit_kd`	1	63–63
`actuated`	1	64–64
`action_effort_gain`	1	65–65
`shape_type`	12	66–77
`shape_transform`	7	78–84
`shape_scale`	3	85–87
`shape_flags`	3	88–90
`shape_is_solid`	1	91–91
`shape_margin`	1	92–92
`shape_gap`	1	93–93
`shape_material_ke`	1	94–94

Field (continued)	Width	Offsets
shape_material_kd	1	95–95
shape_material_kf	1	96–96
shape_material_ka	1	97–97
shape_material_mu	1	98–98
shape_material_restitution	1	99–99
shape_material_mu_torsional	1	100–100
shape_material_mu_rolling	1	101–101
shape_material_kh	1	102–102
world_shape	1	103–103

Dynamic slots 0–12 contain the 13-value body state; slots 13–15 contain joint coordinate, velocity and previous raw action on actuated DOFs; slots 16–21 hold the incoming foot-link wrench; slot 22 marks wrench availability. Other entries remain zero. Directed relation IDs encode none, joint–parent and reverse, joint–child and reverse, DOF–joint and reverse, shape–body and reverse, and contact eligibility. Self and global-token access are supplied by the block.

B Source prototypes and split meaning

All numbers below are multipliers relative to the stock recipe for that leg count. A value omitted in the implementation is one. Density changes preserve the corresponding inertia relationship in the model construction.

Prototype	Legs	Size	Torso	Limbs	Motor	Damping
ant_3	3	1.0	1.0	1.0	1.0	1.0
ant_4	4	1.0	1.0	1.0	1.0	1.0
ant_3_small	3	0.9	1.0	1.0	0.8	1.0
ant_4_large	4	1.1	1.0	1.0	1.2	1.0
ant_3_payload	3	1.0	1.5	1.0	1.0	1.0
ant_4_payload	4	1.0	1.5	1.0	1.0	1.0
ant_3_heavy_limbs	3	1.0	1.0	1.5	1.0	1.5
ant_4_light_limbs	4	1.0	1.0	0.75	1.0	0.75

Table 12: The eight-prototype PPO population. “Torso” and “Limbs” are density multipliers. The successful distillation study uses only the first two rows.

The development split is the stock five-leg variant. Two reserved five-leg parameter variants exist: size 0.95, torso density 1.25 and motor 0.9; and size 1.05, limb density 1.25 and damping 1.25. They were not evaluated in the imitation diagnostic and are too narrow to establish broad structural transfer on their own. Being reserved in the code is not equivalent to supplying a large independent family-level test.

C Implementation and artifact map

The implementation files live under `source/isaaclab_tasks/isaaclab_tasks/contrib/multi_robot_locomotion/`:

`mechanical_observations.py`
Static schema, Newton export, relations, dynamic encoding and action-node mapping.

`mechanical_policy.py`
`MechanicalActor`, attention and message blocks, shared DOF decoder, PPO exploration and prototype critics.

`mechanical_env.py`
Body variants, richer simulator observations and the mechanics/task connection.

`mechanical_imitation.py`
Body-balanced raw-action loss and comparison of physical descriptions.

The collection/fitting/evaluation entry point is `scripts/tools/distill_mechanical_ant.py`. The source research narratives are `MULTI_ROBOT_TRAINING_ARCHITECTURE.md` and `CROSS_EMBODIMENT_RESEARCH_PLAN.md` at the repository root.

C.1 Selected checkpoints and experiment records

Paths below are relative to the IsaacLab repository. They identify local artifacts; a PDF does not archive the checkpoints or training datasets.

- Working attention checkpoint: `logs/shared_ant_graph/imitation_20261007/attention_joint/best.pt`.
- Graph comparison checkpoint: `logs/shared_ant_graph/imitation_20261007/graph_joint/best.pt`.
- Teacher checkpoint: `logs/shared_ant_graph/mechanical_pilot_20261002/controls/specialists_canonical_seed42/final.pt`.
- Imitation results, protocol, status and validation: `logs/shared_ant_graph/imitation_20261007/`.
- Earlier graph/attention PPO comparison: `logs/shared_ant_graph/message_passing_20261006/comparison.json`.
- Inertial ablations and specialist controls: `logs/shared_ant_graph/mechanical_pilot_20261002/controls/summary.json`.
- Earlier joint-only graph continuation: `logs/shared_ant_graph/source_diagnostic_20261002/`.
- Go2 teacher, solo/shared fits, frozen evaluations and diagnostics: `logs/shared_ant_graph/go2_learning_20261007/`.

Older policy videos exist under `logs/shared_ant_graph/mechanical_pilot_20261002/videos/`. They depict the earlier PPO actors, not the new distilled policy, and should not be used as visual evidence for Table 4. New frozen-attention and standing-check recordings accompany the transfer extension under `logs/shared_ant_graph/other_robots_20261007/videos/`; its `index.html` gallery states the control conversion and metric definitions. All displayed trajectories use seed 123, world 0, with resets retained.

D Rebuilding and provenance

The report folder contains the LaTeX source, BibTeX bibliography, vector figures, generated table fragments, and JSON result/protocol snapshots. The PDF can be rebuilt without Isaac Sim, Newton simulation or checkpoint loading. From the IsaacLab repository root, run:

```
uv run --no-sync python \
  research_reports/cross_embodiment/build_report.py
```

The script requires NumPy, Matplotlib, `latexmk`, pdfLaTeX and the standard LaTeX packages used in the preamble. It regenerates the quantitative figures and tables from included snapshots, builds in `.build/`, and places the PDF next to the source. With generated assets already present, `latexmk -pdf` on `cross_embodiment_report.tex` is sufficient from this folder.

`data/provenance.json` records original artifact paths, SHA-256 hashes, branch and Git HEAD. Because the research implementation includes uncommitted files, HEAD alone does not identify the source state. The provenance snapshot is an audit aid, not a complete training-environment archive. Result JSONs included with the report support the quantitative tables; reproducing training additionally requires the assets, environment, checkpoints and full experiment manifests.

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